LAB 11

Lab Title: Strings in C++

1. OBJECTIVE

The purpose of this lab is to understand and apply string handling in C++, including C-style strings (character arrays terminated with a null character \0), string objects from the std::string class, and built-in string manipulation functions such as strlen() and strcat().

2. THEORY

A string is a collection of characters. In C++ there are two commonly used types of strings:

- C-strings (C-style strings) — arrays of type char terminated with the null character \0. The string "Hello" has five characters but occupies six bytes because \0 is appended automatically. The cin extraction operator (>>) stops reading at whitespace; cin.get() and getline() read full lines including spaces.
- String objects — instances of the std::string class. Unlike char arrays, string objects have no fixed length and can be extended as required. The getline() function is used to read a full line into a string object.

Commonly used built-in string functions for C-strings:

- strcpy(s1, s2) — copies string s2 into string s1.
- strcat(s1, s2) — concatenates string s2 onto the end of string s1.
- strlen(s) — returns the length of string s, excluding the null terminator.
- strcmp(s1, s2) — returns 0 if the two strings are equal.
- strlwr(s) / strupr(s) — converts a string to lowercase or uppercase.

3. SOFTWARE USED

- DEV-C++ IDE
- C++ Compiler (bundled with DEV-C++)

4. PROCEDURE

Task 1 — String Length Calculation

Write a C++ program that declares and initializes a string variable called message with a phrase of your choice. Use a built-in function to calculate and display the length of the string.

```
#include <iostream>
#include <cstring>
using namespace std;

int main()
{
    char message[] = "Electrical Engineering";  // initialize string
    int len = strlen(message);              // calculate length
    cout << "String: " << message << endl;  // display string
    cout << "Length: " << len    << endl;  // display length
    return 0;
}
```

Output:

```
String: Electrical Engineering
Length: 25
```

Task 2 — String Concatenation

Declare two string variables called firstName and lastName and initialize them with your first and last name. Concatenate the two strings to form a full name and display it.

```
#include <iostream>
#include <cstring>
using namespace std;
```

```
int main()
{
    char firstName[50] = "Talha";           // first name
    char lastName[50]  = " Ali";            // last name (leading space)
    strcat(firstName, lastName);            // concatenate strings
    cout << "Full Name: " << firstName << endl;  // display result
    return 0;
}
```

Output:

```
Full Name: Talha Ali
```

5. CONCLUSION

This lab provided practical experience with string handling in C++. C-style strings were declared and initialized as character arrays. The strlen() function was used to determine string length and the strcat() function was used to concatenate two strings. The distinction between cin >> (which stops at whitespace) and cin.get() or getline() (which read full lines) was clearly understood. The null terminator \0 was confirmed as an essential part of every C-style string.